

Pangolin Wildlife Trade Spatial Modeling - Gabon

Objective 3: Random Forest Model Evaluation and Feature Importance Analysis

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1. Setup, File Existence Check, and Raster Stack Loading

```
# =====
# CHUNK 1: ENVIRONMENT SETUP & RASTER COVARIATE IMPORT
# =====

library(terra)
library(sf)
library(dplyr)

# Set base path for covariates
cov_dir <- "C:/Users/eldayit/OneDrive - University of Maryland/Desktop/All files/Ventures/Pangolins_Gabon"

# Explicit covariate filenames
cov_files <- c(
  "B_distance_to_Forest_Concessions.tif",
  "B_distance_to_InternationalGabonBorders.tif",
  "B_distance_to_MajorGabonCities.tif",
  "B_distance_to_majorRoads.tif",
  "B_distance_to_ProtectedAreas.tif",
  "B_distance_to_waterways.tif",
  "B_LAI_2024.tif",
  "B_Population_2020.tif",
  "B_SRTM_2000.tif"
)

# Construct full paths and verify file existence
cov_paths <- file.path(cov_dir, cov_files)
missing_files <- cov_paths[!file.exists(cov_paths)]

if (length(missing_files) > 0) {
  stop("The following raster files were not found:\n", paste(missing_files, collapse = "\n"))
}

# Load rasters into SpatRaster stack and clean layer names
covariates_raw <- rast(cov_paths)
names(covariates_raw) <- gsub("\\.tif$", "", cov_files)

cat("Chunk 1 Complete: All 9 raw raster covariates successfully imported.\n")

## Chunk 1 Complete: All 9 raw raster covariates successfully imported.
```

2. Spatial Coherence Check, Resampling Alignment, & Composite Index

```
# =====  
# CHUNK 2: SPATIAL ALIGNMENT & COMPOSITE COVARIATE CALCULATIONS  
# =====  
  
library(ggplot2)  
library(tidyterra)  
  
# 1. Align all rasters to the template (first layer) geometry  
template_raster <- covariates_raw[[1]]  
aligned_list    <- list()  
  
for (i in 1:nlyr(covariates_raw)) {  
  lyr <- covariates_raw[[i]]  
  if (!compareGeom(template_raster, lyr, stopOnError = FALSE)) {  
    lyr <- resample(lyr, template_raster, method = "bilinear")  
  }  
  aligned_list[[i]] <- lyr  
}  
  
covariates <- rast(aligned_list)  
  
# 2. Compute anthropogenic composite accessibility index  
covariates$B_city_road_composite <- covariates[["B_distance_to_MajorGabonCities"]] +  
  covariates[["B_distance_to_majorRoads"]]  
  
# 3. Visualize component vs. composite accessibility covariates  
accessibility_stack <- subset(  
  covariates,  
  c("B_distance_to_MajorGabonCities",  
    "B_distance_to_majorRoads",  
    "B_city_road_composite")  
)  
  
ggplot() +  
  geom_spatraster(data = accessibility_stack) +  
  facet_wrap(~lyr, ncol = 3) +  
  scale_fill_viridis_c(  
    option = "magma",  
    name    = "Distance (m)",  
    na.value = "transparent"  
  ) +  
  labs(  
    title    = "Anthropogenic Accessibility Covariates (Gabon)",  
    subtitle = "Individual Components vs. Composite Accessibility Index"  
  ) +  
  theme_minimal(base_size = 11) +  
  theme(  
    axis.text    = element_blank(),  
    axis.ticks   = element_blank(),  
    panel.grid   = element_blank(),  
    strip.text   = element_text(face = "bold", size = 10),  
    plot.title   = element_text(face = "bold")  
  )
```

)

Anthropogenic Accessibility Covariates (Gabon)

Individual Components vs. Composite Accessibility Index

Distance to Major Gabon Cities Distance to major Roads City road composite



3. Complete Random Forest Modeling & Spatial Prediction Pipeline Function

```
# =====  
# CHUNK 3: RANDOM FOREST MODEL EVALUATION & FEATURE IMPORTANCE FUNCTION  
# =====  
  
library(ranger)  
library(pROC)  
library(spdep)  
library(cowplot)  
library(scales)  
  
# Clean, publication-ready variable lookup labels  
variable_labels <- c(  
  "B_distance_to_MajorGabonCities" = "Dist. major cities",  
  "B_city_road_composite" = "Composite dist. major cities & roads",  
  "B_distance_to_InternationalGabonBorders" = "Dist. international borders",  
  "B_distance_to_majorRoads" = "Dist. major roads",  
  "B_distance_to_Forest_Concessions" = "Dist. forest concessions",  
  "B_LAI_2024" = "Leaf area index",  
  "B_Population_2020" = "Population density",  
  "B_SRTM_2000" = "Elevation",  
  "B_distance_to_ProtectedAreas" = "Dist. protected areas",
```

```

"B_distance_to_waterways"          = "Dist. waterways"
)

# Define output directory for saving plots
output_dir <- "C:/Users/eldayit/OneDrive - University of Maryland/Desktop/All files/Ventures/Pangolins (
if (!dir.exists(output_dir)) dir.create(output_dir, recursive = TRUE)

#' Run Random Forest Evaluation & Feature Importance for a Given Crime Stage
run_crime_stage_rf <- function(stage_path, stage_name, covariates_stack) {

  # -----
  # 1. Load Presences & Extract Environmental Values
  # -----
  pts <- st_read(stage_path, quiet = TRUE) %>%
    st_transform(crs(covariates_stack))

  pres_env <- terra::extract(covariates_stack, pts, ID = FALSE)
  valid_pres <- complete.cases(pres_env)

  pres_df <- data.frame(presence = 1, pres_env[valid_pres, ])
  pts_valid <- pts[valid_pres, ]
  n_pres <- nrow(pres_df)

  # -----
  # 2. Sample Sampling-Bias Weighted Background Points (2:1 Ratio)
  # -----
  road_dist <- covariates_stack[["B_distance_to_majorRoads"]]
  bias_surface <- exp(-0.0001 * road_dist)

  set.seed(42)
  bg_coords <- terra::spatSample(
    bias_surface,
    size = n_pres * 2,
    method = "weights",
    xy = TRUE,
    na.rm = TRUE
  )

  bg_pts <- st_as_sf(as.data.frame(bg_coords), coords = c("x", "y"), crs = crs(covariates_stack))
  bg_env <- terra::extract(covariates_stack, bg_pts, ID = FALSE)
  valid_bg <- complete.cases(bg_env)

  bg_df <- data.frame(presence = 0, bg_env[valid_bg, ])
  bg_pts_valid <- bg_pts[valid_bg, ]

  # Combine presence and background datasets
  df <- rbind(pres_df, bg_df)
  df$presence <- as.factor(df$presence)

  # -----
  # 3. Train/Test Split (70/30) & Random Forest Fitting
  # -----

```

```

set.seed(123)
train_idx <- sample(seq_len(nrow(df)), size = 0.70 * nrow(df))
train_data <- df[train_idx, ]
test_data <- df[-train_idx, ]

rf_model <- ranger(
  formula = presence ~ .,
  data = train_data,
  probability = TRUE,
  num.trees = 700,
  importance = "permutation"
)

# Evaluate Test AUC
test_preds <- predict(rf_model, data = test_data)$predictions[, "1"]
roc_obj <- roc(test_data$presence, test_preds, quiet = TRUE)

# -----
# 4. Residual Spatial Autocorrelation (Moran's I)
# -----
full_preds <- predict(rf_model, data = df)$predictions[, "1"]
obs_numeric <- as.numeric(as.character(df$presence))
residuals_vec <- obs_numeric - full_preds

all_geom <- c(st_geometry(pts_valid), st_geometry(bg_pts_valid))
coords_mat <- st_coordinates(all_geom)

nb <- knn2nb(knearneigh(coords_mat, k = 8))
lw <- nb2listw(nb, style = "W")

moran_res <- moran.test(residuals_vec, lw)

# -----
# 5. Print Goodness of Fit & Performance Summary
# -----
cat("\n===== \n")
cat("MODEL EVALUATION SUMMARY:", stage_name, "\n")
cat("===== \n")
cat(sprintf("Sample Sizes : %d Presences | %d Background Points\n", n_pres, nrow(bg_df)))
cat(sprintf("Test AUC Score : %.4f\n", auc(roc_obj)))
cat(sprintf("OOB Error Rate : %.4f\n", rf_model$prediction.error))
cat(sprintf("Moran's I (Res): %.4f (p-value: %.4f)\n",
  moran_res$estimate[["Moran I statistic"]], moran_res$p.value))
cat("===== \n\n")

# -----
# 6. Publication-Ready Feature Importance Plot
# -----
imp_scores <- importance(rf_model)
imp_df <- data.frame(
  Variable = names(imp_scores),
  Importance = as.numeric(imp_scores)
) %>% arrange(desc(Importance))

```

```

p_imp <- ggplot(imp_df, aes(x = reorder(Variable, Importance), y = Importance)) +
  geom_col(fill = "#2b5c8f", width = 0.7) +
  geom_text(
    aes(label = sprintf("%.2f%%", Importance * 100)),
    hjust = -0.15, size = 3.5, fontface = "bold", color = "black"
  ) +
  scale_x_discrete(labels = variable_labels) +
  scale_y_continuous(
    limits = c(0, max(imp_df$Importance) * 1.25),
    labels = percent_format(accuracy = 1),
    expand = c(0, 0)
  ) +
  coord_flip() +
  labs(
    title = paste0("Feature Importance - ", stage_name),
    x      = "Predictor Variables",
    y      = "Mean Decrease in Accuracy (%)"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    panel.grid.minor = element_blank(),
    plot.title       = element_text(face = "bold", size = 13),
    axis.title.x     = element_text(face = "bold", size = 11, margin = margin(t = 10)),
    axis.title.y     = element_text(face = "bold", size = 11, margin = margin(r = 10)),
    axis.text        = element_text(face = "bold", color = "black")
  )

# Display plot in R Markdown output
print(p_imp)

# Save Importance Plot PNG
clean_stage_id <- gsub("[^A-Za-z0-9]", "_", stage_name)
ggsave(
  filename = file.path(output_dir, paste0("Feature_Importance_", clean_stage_id, ".png")),
  plot      = p_imp,
  width     = 9, height = 5.5, dpi = 300, bg = "white"
)

return(list(
  model      = rf_model,
  roc        = roc_obj,
  moran      = moran_res,
  importance = imp_df
))
}

```

4. EXECUTE MODELING PIPELINE FOR ALL CRIME STAGES

```

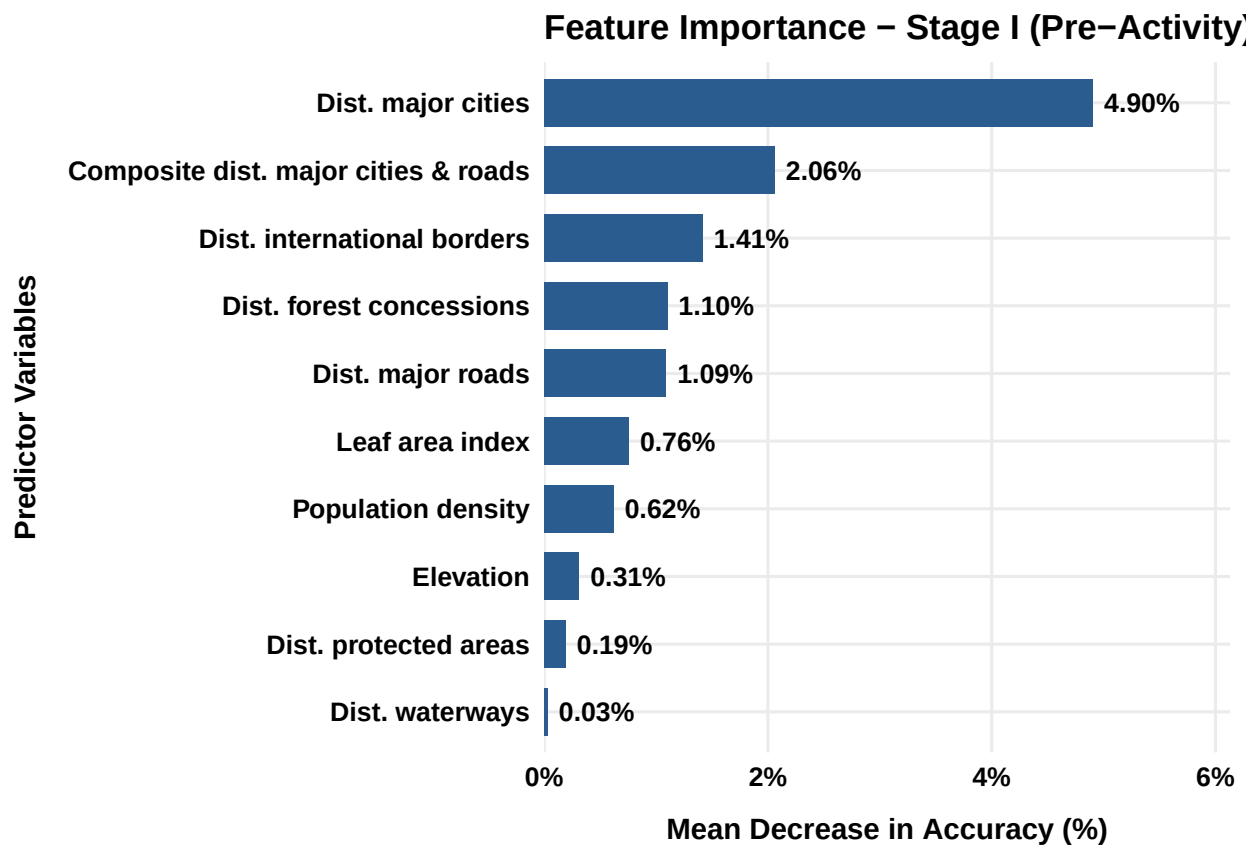
# =====
# CHUNK 4: EXECUTE MODELING EVALUATION & IMPORTANCE FOR ALL STAGES
# =====

input_dir_obj3 <- "C:/Users/eldayit/OneDrive - University of Maryland/Desktop/All files/Ventures/Pangol

```

```
# Stage I (Pre-Activity)
stage1_results <- run_crime_stage_rf(
  stage_path      = file.path(input_dir_obj3, "Crime_stage_I.gpkg"),
  stage_name      = "Stage I (Pre-Activity)",
  covariates_stack = covariates
)
```

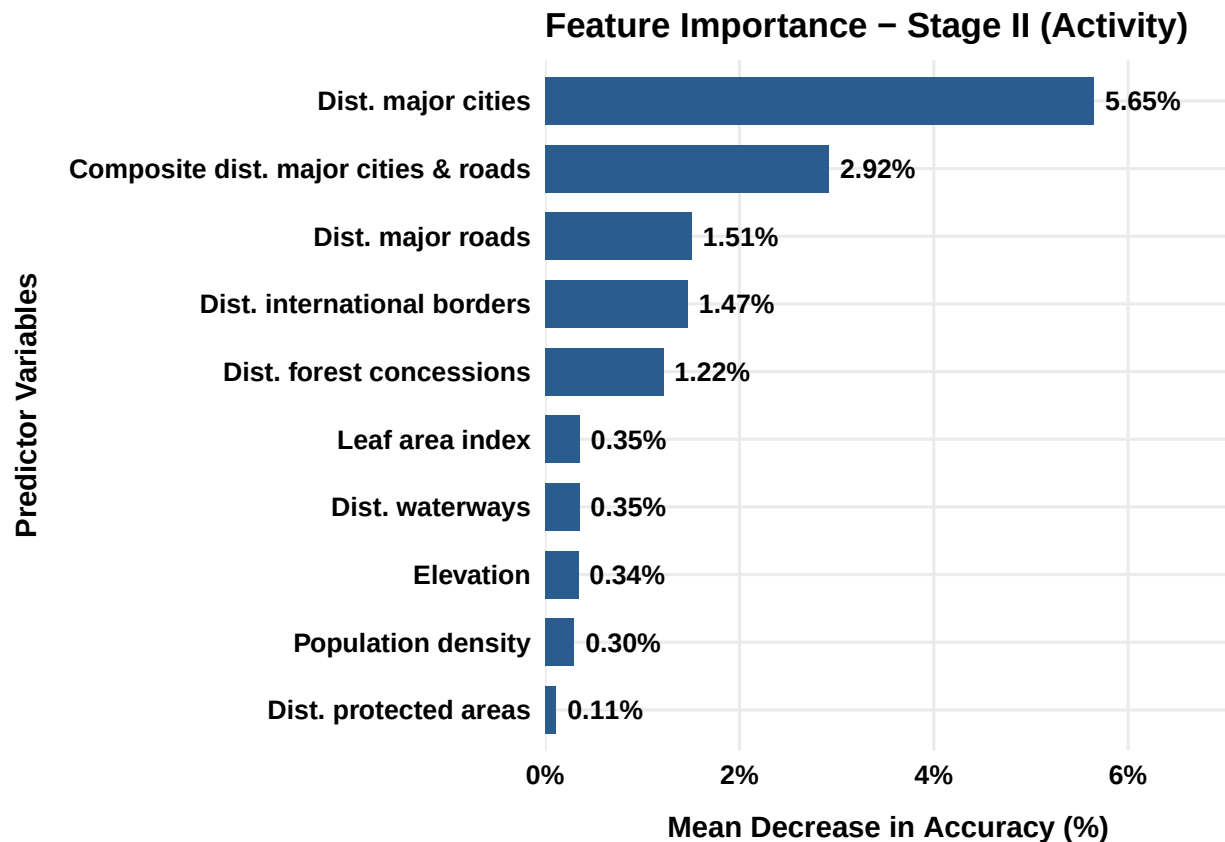
```
##
## =====
## MODEL EVALUATION SUMMARY: Stage I (Pre-Activity)
## =====
## Sample Sizes   : 676 Presences | 1262 Background Points
## Test AUC Score : 0.6953
## OOB Error Rate : 0.1884
## Moran's I (Res): -0.0032 (p-value: 0.5987)
## =====
```



```
# Stage II (Activity)
stage2_results <- run_crime_stage_rf(
  stage_path      = file.path(input_dir_obj3, "Crime_stage_II.gpkg"),
  stage_name      = "Stage II (Activity)",
  covariates_stack = covariates
)
```

```
##
```

```
## =====
## MODEL EVALUATION SUMMARY: Stage II (Activity)
## =====
## Sample Sizes   : 745 Presences | 1395 Background Points
## Test AUC Score : 0.7548
## OOB Error Rate : 0.1901
## Moran's I (Res): -0.0069 (p-value: 0.7363)
## =====
```



```
# Stage III (Post-Activity)
stage3_results <- run_crime_stage_rf(
  stage_path      = file.path(input_dir_obj3, "Crime_stage_III.gpkg"),
  stage_name      = "Stage III (Post-Activity)",
  covariates_stack = covariates
)
```

```
##
## =====
## MODEL EVALUATION SUMMARY: Stage III (Post-Activity)
## =====
## Sample Sizes   : 675 Presences | 1260 Background Points
## Test AUC Score : 0.7458
## OOB Error Rate : 0.1829
## Moran's I (Res): -0.0089 (p-value: 0.7802)
## =====
```